

## 19—NATURAL RESOURCES

| REF NO                                                                                       | DESCRIPTION                                                                                             | SYMBOL | CARTOGRAPHIC SPECIFICATIONS*                                                                               | NOTES ON USAGE*                                                                                                                                                 |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>19.1—Veins and mineralized areas; mineral resource areas; metamorphic facies boundary</b> |                                                                                                         |        |                                                                                                            |                                                                                                                                                                 |
| 19.1.1                                                                                       | Vein, veinlet, or mineralized stringer—Identity and existence certain, location accurate                |        | lineweight .25 mm color 100% red<br>→ 8.0 mm ←<br>H-8<br>→ .75 mm ←<br>dot diameter .75 mm; spacing 4.5 mm | May also be shown in black or other colors.                                                                                                                     |
| 19.1.2                                                                                       | Vein, veinlet, or mineralized stringer—Identity or existence questionable, location accurate            |        |                                                                                                            |                                                                                                                                                                 |
| 19.1.3                                                                                       | Vein, veinlet, or mineralized stringer—Identity and existence certain, location approximate             |        | 3.625 mm<br>→ .75 mm ←<br>→ .75 mm ←                                                                       |                                                                                                                                                                 |
| 19.1.4                                                                                       | Vein, veinlet, or mineralized stringer—Identity or existence questionable, location approximate         |        |                                                                                                            |                                                                                                                                                                 |
| 19.1.5                                                                                       | Vein, veinlet, or mineralized stringer—Identity and existence certain, location concealed               |        | .5 mm<br>→ .75 mm ←<br>→ .75 mm ←                                                                          |                                                                                                                                                                 |
| 19.1.6                                                                                       | Vein, veinlet, or mineralized stringer—Identity or existence questionable, location concealed           |        |                                                                                                            |                                                                                                                                                                 |
| 19.1.7                                                                                       | Vein, veinlet, or mineralized stringer—Showing type of mineral occurrence                               |        | Cu ← H-8 (100% black)                                                                                      |                                                                                                                                                                 |
| 19.1.8                                                                                       | Inclined vein, veinlet, or mineralized stringer (1st option)—Showing dip value and direction            |        | tick length 1.75 mm; lineweight .2 mm<br>35 ← H-6 (100% black)<br>tick color 100% black                    | Place tick, arrow, or other line-symbol decoration where observation was made.<br>Add arrowhead or '90' to ticks showing dip if necessary for clarity.          |
| 19.1.9                                                                                       | Inclined vein, veinlet, or mineralized stringer (2nd option)—Showing dip value and direction            |        | tick length 1.375 mm; lineweight .2 mm<br>15 ← .875 mm<br>30°                                              |                                                                                                                                                                 |
| 19.1.10                                                                                      | Vertical or near-vertical vein, veinlet, or mineralized stringer (1st option)                           |        | tick length 2.5 mm; lineweight .2 mm                                                                       |                                                                                                                                                                 |
| 19.1.11                                                                                      | Vertical or near-vertical vein, veinlet, or mineralized stringer (2nd option)                           |        | 90 ← H-6 (100% black)                                                                                      |                                                                                                                                                                 |
| 19.1.12                                                                                      | Small, minor inclined vein, veinlet, or mineralized stringer—Showing strike and dip                     |        | H-6 (100% black) → 70<br>1.25 mm<br>5.5 mm<br>lineweight .25 mm; line color 100% red                       | May also be shown in black or other colors.                                                                                                                     |
| 19.1.13                                                                                      | Small, minor vertical or near-vertical vein, veinlet, or mineralized stringer—Showing strike            |        | 2.5 mm                                                                                                     |                                                                                                                                                                 |
| 19.1.14                                                                                      | Zone of mineralized or altered rock (1st option)                                                        |        | pattern 405-R (at 45°)                                                                                     | Add labels to show specific types of alteration. May be used alone or may overprint other mapped units. May also be shown in black or other colors.             |
| 19.1.15                                                                                      | Zone of mineralized or altered rock (2nd option)                                                        |        | pattern 405-R in 50% red (at 45°)                                                                          |                                                                                                                                                                 |
| 19.1.16                                                                                      | Zone of mineralized or altered rock, showing high level of mineralization                               |        | pattern 119-R                                                                                              |                                                                                                                                                                 |
| 19.1.17                                                                                      | Zone of mineralized or altered rock, showing low level of mineralization                                |        | pattern 117-R                                                                                              |                                                                                                                                                                 |
| 19.1.18                                                                                      | Area of identified resources                                                                            |        | lineweight .5 mm<br>color 100% red                                                                         | Usually reserved for use on special-purpose maps, not on general-purpose geologic maps. Generally shown in red, but may also be shown in black or other colors. |
| 19.1.19                                                                                      | Area of high mineral resource potential                                                                 |        | lineweight .3 mm<br>line and text color 100% red<br>H-12<br>pattern 427-R in 50% red                       |                                                                                                                                                                 |
| 19.1.20                                                                                      | Area of moderate mineral resource potential                                                             |        | lineweight .3 mm<br>H-12<br>pattern 229-R (at 45°) in 50% red                                              |                                                                                                                                                                 |
| 19.1.21                                                                                      | Area of low mineral resource potential                                                                  |        | lineweight .2 mm<br>H-10                                                                                   |                                                                                                                                                                 |
| 19.1.22                                                                                      | Area considered to have mineral resource potential but not evaluated, mostly because of inadequate data |        | lineweight .2 mm<br>H-10<br>dash 1.75 mm; space .5 mm                                                      |                                                                                                                                                                 |
| 19.1.23                                                                                      | Metamorphic facies boundary—Showing approximate boundary between diagnostic mineral assemblages         |        | H-8<br>Greenschist<br>Amphibolite<br>dot diameter .5 mm; spacing .5 mm<br>line and text color 100% red     | May also be shown in black or other colors.                                                                                                                     |

\*For more information, see general guidelines on pages A-i to A-v.